

SAFETY DATA SHEET



RESURIS® SC

Version	Revision Date:	SDS Number:	Date of last issue: -
2.0	05.02.2026	50002641	Date of first issue: 16.12.2024

SECTION 1. IDENTIFICATION

Product name : RESURIS® SC

Manufacturer or supplier's details

Company : FMC QUÍMICA DO BRASIL LTDA.

Address : AVENIDA DR. JOSÉ BONIFÁCIO
COUTINHO NOGUEIRA 150 - 1º
ANDAR - JARDIM MADALENA,
CAMPINAS SP BRASIL
TELEFONE: (19) 2042.4500

Registrant : FMC QUÍMICA S.A.
CARLOS PELLEGRINI 719, PISO 9, CABA
ARGENTINA

Emergency telephone : Argentina: 54-1159839431 (CHEMTREC)
All other countries: +1 651 / 632-6793 (Collect)

Medical Emergency Number : FMC (General) - (011) 5984-3700
Hospital Nacional Prof. Alejandro Posadas, Centro Nacional de
Intoxicaciones. (Toxicológica) - 0800- 333 -0160 / (011)4658-
7777 / (011) 4654-6648
Hospital de Niños Ricardo Gutierrez, Unidad de Toxicología.
(Toxicológica) - 0800-444-8694 / (011)4962-6666 / (011)4962-
2247
Hospital General de Agudos J. A. Fernández ,Unidad de Toxi-
cología. (Toxicológica) - (011) 4808-2655 / (011)4808-2606
TAS ,Toxicología , Asesoramiento y Servicios. (Toxicológica) -
0800-888-8694 / (0341) 4242727
Bomberos (General) – 100
Policia (General) – 101 – 911
Defensa Civil (General) – 103
Emergencias médicas (General) – 107

Recommended use of the chemical and restrictions on use

Recommended use : Fungicide

Restrictions on use : Use as recommended by the label.

SECTION 2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity (Oral) : Category 5

Acute toxicity (Inhalation) : Category 2

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Acute toxicity (Dermal)	:	Category 5
Skin sensitization	:	Category 1
Carcinogenicity	:	Category 2
Specific target organ toxicity - single exposure	:	Category 3 (Respiratory system)
Short-term (acute) aquatic hazard	:	Category 1
Long-term (chronic) aquatic hazard	:	Category 1

GHS label elements

Hazard pictograms :



Signal Word : DANGER

Hazard Statements : H303 + H313 May be harmful if swallowed or in contact with skin.
H317 May cause an allergic skin reaction.
H330 Fatal if inhaled.
H335 May cause respiratory irritation.
H351 Suspected of causing cancer.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P260 Do not breathe mist or vapors.
P271 Use only outdoors or in a well-ventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.
P284 Wear respiratory protection.

Response:
P302 + P352 IF ON SKIN: Wash with plenty of water.
P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.
P312 Call a POISON CENTER/ doctor if you feel unwell.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before

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reuse.
P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Chemical name	CAS-No.	Concentration (% w/w)
chlorothalonil (ISO)	1897-45-6	>= 30 -< 50
Fluindapyr	1383809-87-7	>= 2,5 -< 5
1,2-benzisothiazol-3(2H)-one	2634-33-5	>= 0,025 -< 0,1

SECTION 4. FIRST AID MEASURES

- General advice : Move out of dangerous area.
Consult a physician.
Show this material safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended.
- If inhaled : Call a physician or poison control center immediately.
If unconscious, place in recovery position and seek medical advice.
- In case of skin contact : Wash off with soap and water.
If symptoms persist, call a physician.
Wash contaminated clothing before re-use.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Do not induce vomiting without medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

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Take victim immediately to hospital.

Most important symptoms and effects, both acute and delayed : May be harmful if swallowed or in contact with skin.
May cause an allergic skin reaction.
Fatal if inhaled.
May cause respiratory irritation.
Suspected of causing cancer.
Exposure to skin may result in mild symptoms include itching, hives or rash, and skin redness. More severe symptoms include sneezing, itchy watery eyes, and difficulty breathing.

Protection of first-aiders : Avoid inhalation, ingestion and contact with skin and eyes.

Notes to physician : Treat symptomatically.

SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.

Unsuitable extinguishing media : Do not spread spilled material with high-pressure water streams.

Specific hazards during fire fighting : Do not allow run-off from fire fighting to enter drains or water courses.

Hazardous combustion products : Fire may produce irritating, corrosive and/or toxic gases.
Nitrogen oxides (NO_x)
Hydrogen cyanide
Chlorinated compounds
Hydrogen chloride
Carbon oxides
Sulfur oxides
Fluorine compounds

Specific extinguishing methods : Remove undamaged containers from fire area if it is safe to do so.
Use a water spray to cool fully closed containers.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment for fire-fighters : Firefighters should wear protective clothing and self-contained breathing apparatus.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Evacuate personnel to safe areas.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.

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Ensure adequate ventilation.

- Environmental precautions : Prevent further leakage or spillage if safe to do so.
Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.
- Methods and materials for containment and cleaning up : Never return spills in original containers for re-use.
Collect as much of the spill as possible with a suitable absorbent material.
Pick up and transfer to properly labeled containers.
Keep in suitable, closed containers for disposal.

SECTION 7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
- Conditions for safe storage : Prevent unauthorized access.
Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.
- Further information on storage stability : No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Ingredients with workplace control parameters

Contains no substances with occupational exposure limit values.

Personal protective equipment

- Respiratory protection : In the case of dust or aerosol formation use respirator with an

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approved filter.

Hand protection
Material : Protective gloves

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Plan first aid action before beginning work with this product.

Hygiene measures : Avoid contact with skin, eyes and clothing.
Do not inhale aerosol.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Color : beige

Odor : characteristic

Odor Threshold : No data available

pH : 5,1 (ca. 20 °C)
Concentration: 10 g/l

Melting point/ range : No data available

Boiling point/boiling range : 89,8 °C

Flash point : No flash up to boiling point.

Evaporation rate : No data available

Self-ignition : No data available

Upper explosion limit / Upper flammability limit : No data available

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Lower explosion limit / Lower flammability limit	:	No data available
Vapor pressure	:	No data available
Relative vapor density	:	No data available
Relative density	:	No data available
Density	:	ca. 1,25 g/cm ³ (ca. 20 °C) Method: OECD Test Guideline 109
Solubility(ies) Water solubility	:	No data available
Partition coefficient: n-octanol/water	:	No data available
Autoignition temperature	:	No data available
Decomposition temperature	:	No data available
Viscosity Viscosity, dynamic	:	ca. 2.535 mPa.s (ca. 20 °C) Method: OECD Test Guideline 114 GLP: yes ca. 1.552 mPa.s (ca. 40 °C) Method: OECD Test Guideline 114 GLP: yes
Viscosity, kinematic	:	No data available
Explosive properties	:	Not explosive
Oxidizing properties	:	Non-oxidizing
Surface tension	:	49,54 mN/m, 1,066 g/l, OECD Test Guideline 115
Molecular weight	:	Not applicable
Metal corrosion rate	:	Not corrosive to metals.

SECTION 10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	No decomposition if stored and applied as directed.
Possibility of hazardous reac-	:	No decomposition if stored and applied as directed.

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Conditions to avoid	:	Avoid extreme temperatures. Avoid formation of aerosol.
Incompatible materials	:	Avoid strong acids, bases, and oxidizers.
Hazardous decomposition products	:	No hazardous decomposition products are known.

SECTION 11. TOXICOLOGICAL INFORMATION**Acute toxicity**

May be harmful if swallowed or in contact with skin.
Fatal if inhaled.

Product:

Acute oral toxicity	:	(Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 423 GLP: yes Assessment: The component/mixture is minimally toxic after single ingestion. Remarks: no mortality
Acute inhalation toxicity	:	LC50 (Rat, male): 0,48 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Symptoms: Fatality, pulmonary hemorrhage, Tremors, chromodacryorrhea GLP: yes
Acute dermal toxicity	:	(Rat, female): > 2.000 mg/kg Method: OECD Test Guideline 402 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin. Remarks: no mortality

Components:**chlorothalonil (ISO):**

Acute oral toxicity	:	LD50 (Rat): > 5.000 mg/kg
Acute inhalation toxicity	:	LC50 (Rat): 0,1 mg/l Exposure time: 4 h Test atmosphere: dust/mist
Acute dermal toxicity	:	LD50 (Rat): > 2.000 mg/kg

Fluindapyr:

Acute oral toxicity	:	LD50 (Rat, female): > 2.000 mg/kg
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Method: OECD Test Guideline 423
GLP: yes
Assessment: The component/mixture is minimally toxic after single ingestion.
Remarks: no mortality

Acute inhalation toxicity : LC50 (Rat, male and female): > 5,19 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: ataxia, Breathing difficulties
GLP: yes
Remarks: no mortality

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Symptoms: Irritation
GLP: yes
Assessment: The component/mixture is minimally toxic after single contact with skin.
Remarks: no mortality

1,2-benzisothiazol-3(2H)-one:

Acute dermal toxicity : LD50 (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species : reconstructed human epidermis (RhE)
Method : OECD Test Guideline 431
Result : Not corrosive
GLP : yes

Species : reconstructed human epidermis (RhE)
Assessment : No skin irritation
Method : OECD Test Guideline 439
GLP : yes

Components:

chlorothalonil (ISO):

Species : Rabbit
Result : Mild skin irritant

Fluindapyr:

Species : Rabbit
Assessment : Not classified as irritant
Method : OECD Test Guideline 404

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GLP : yes

Assessment : Not classified as irritant
Method : EPISKIN Human Skin Model Test

1,2-benzisothiazol-3(2H)-one:

Species : Rabbit
Exposure time : 72 h
Method : OECD Test Guideline 404
Result : No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Species : Bovine cornea
Result : not corrosive
Method : OECD Test Guideline 437

Remarks : Vapors may cause irritation to the eyes, respiratory system and the skin.

Components:

chlorothalonil (ISO):

Species : Rabbit
Result : Irreversible effects on the eye

Fluindapyr:

Species : Rabbit
Result : No eye irritation
Method : OECD Test Guideline 405
GLP : yes

Result : not corrosive
Method : Bovine cornea (BCOP)
GLP : yes

1,2-benzisothiazol-3(2H)-one:

Species : Bovine cornea
Result : No eye irritation
Method : OECD Test Guideline 437

Species : Rabbit
Result : Irreversible effects on the eye
Method : EPA OPP 81-4

Respiratory or skin sensitization

Skin sensitization

May cause an allergic skin reaction.

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Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Test Type	: Local lymph node assay (LLNA)
Routes of exposure	: Dermal
Species	: mice
Method	: OECD Test Guideline 429
Result	: May cause sensitization by skin contact.
GLP	: yes

Components:**chlorothalonil (ISO):**

Result	: May cause sensitization by skin contact.
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Fluindapyr:

Test Type	: Local lymph node assay (LLNA)
Routes of exposure	: Skin contact
Method	: OECD Test Guideline 429
Result	: May cause sensitization by skin contact.
GLP	: yes

1,2-benzisothiazol-3(2H)-one:

Test Type	: Maximization Test
Species	: Guinea pig
Method	: OECD Test Guideline 406
Result	: May cause sensitization by skin contact.

Species	: Guinea pig
Method	: FIFRA 81.06
Result	: May cause sensitization by skin contact.

Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Product:

Genotoxicity in vitro	: Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative
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Genotoxicity in vivo	: Test Type: Micronucleus test Species: Mouse Method: OECD Test Guideline 474 Result: negative
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Components:**chlorothalonil (ISO):**

Genotoxicity in vitro	: Test Type: Chromosome aberration test in vitro Result: positive
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Genotoxicity in vivo : Test Type: Micronucleus test
Result: negative

Test Type: chromosome aberration assay
Result: negative

Germ cell mutagenicity - Assessment : Weight of evidence does not support classification as a germ cell mutagen.

Fluindapyr:

Genotoxicity in vitro : Test Type: Chromosome aberration test in vitro
Test system: lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative

Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 490
Result: negative

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes

Genotoxicity in vivo : Test Type: Mammalian bone marrow sister chromatid exchange
Species: Mouse
Result: negative

Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative

1,2-benzisothiazol-3(2H)-one:

Genotoxicity in vitro : Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: Ames test
Method: OECD Test Guideline 471

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Result: negative

Test Type: Chromosome aberration test in vitro

Method: OECD Test Guideline 473

Result: positive

Genotoxicity in vivo

: Test Type: unscheduled DNA synthesis assay

Species: Rat (male)

Cell type: Liver cells

Application Route: Ingestion

Exposure time: 4 h

Method: OECD Test Guideline 486

Result: negative

Test Type: Micronucleus test

Species: Mouse

Application Route: Oral

Method: OECD Test Guideline 474

Result: negative

Germ cell mutagenicity -
Assessment

: Weight of evidence does not support classification as a germ cell mutagen.

Carcinogenicity

Suspected of causing cancer.

Components:**chlorothalonil (ISO):**

Carcinogenicity - Assessment

: Limited evidence of carcinogenicity in animal studies

Fluindapyr:

Species

: Mouse

Application Route

: Oral

Exposure time

: 18 month(s)

Method

: OECD Test Guideline 451

Result

: Not a carcinogenic hazard

Species

: Rat

Application Route

: Oral

Exposure time

: 2 Years

Method

: OECD Test Guideline 453

Result

: Not a carcinogenic hazard

GLP

: yes

Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:**chlorothalonil (ISO):**

Reproductive toxicity - Assessment

: Weight of evidence does not support classification for reproductive toxicity

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Fluindapyr:

Effects on fertility : Test Type: Two-generation study
General Toxicity Parent: NOEL: ca. 100 ppm
Fertility: NOAEL: ca. 400 ppm
Early Embryonic Development: NOAEL: ca. 400 ppm
Method: OECD Test Guideline 416
GLP: yes

1,2-benzisothiazol-3(2H)-one:

Effects on fertility : Species: Rat, male
Application Route: Ingestion
General Toxicity Parent: NOAEL: 18,5 mg/kg body weight
General Toxicity F1: NOAEL: 48 mg/kg body weight
Fertility: NOAEL: 112 mg/kg bw/day
Symptoms: No effects on reproduction parameters.
Method: OPPTS 870.3800
Result: negative

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

May cause respiratory irritation.

Product:

Routes of exposure : Inhalation
Target Organs : Lungs
Assessment : Causes damage to organs.
Remarks : ANVISA assessment

Components:**chlorothalonil (ISO):**

Assessment : May cause respiratory irritation.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Components:**chlorothalonil (ISO):**

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

1,2-benzisothiazol-3(2H)-one:

Assessment : The substance or mixture is not classified as specific target organ toxicant, repeated exposure.

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Repeated dose toxicity

Components:

chlorothalonil (ISO):

Species	: Rat
NOAEL	: 1,8 mg/kg
Application Route	: Oral
Exposure time	: 26 months

Species	: Rat
NOAEL	: 1,5 mg/kg
Application Route	: Oral
Exposure time	: 90 d

Fluindapyr:

Species	: Rat
NOAEL	: 1.000 mg/kg
Application Route	: Dermal
Exposure time	: 21 d
Number of exposures	: 5 d/w for 6 hr
Dose	: 0,100,300,1000 mg/kg bw/d
Method	: OECD Test Guideline 410
GLP	: yes
Symptoms	: Skin irritation

1,2-benzisothiazol-3(2H)-one:

Species	: Rat, male and female
NOAEL	: 15 mg/kg
Application Route	: Ingestion
Exposure time	: 28 d
Method	: OECD Test Guideline 407
Symptoms	: Irritation

Species	: Rat, male and female
NOAEL	: 69 mg/kg
Application Route	: Ingestion
Exposure time	: 90 d
Symptoms	: Irritation, Reduced body weight

Aspiration toxicity

Based on available data, the classification criteria are not met.

Further information

Product:

Remarks	: No data available
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SECTION 12. ECOLOGICAL INFORMATION**Ecotoxicity****Product:**

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|---|---|--|
| Toxicity to fish | : | LC50 (Danio rerio (zebra fish)): 0,017 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
GLP: yes |
| Toxicity to daphnia and other aquatic invertebrates | : | EC50 (Daphnia magna (Water flea)): 0,13 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
GLP: yes |
| Toxicity to algae/aquatic plants | : | EyC50 (Pseudokirchneriella subcapitata (algae)): 2,07 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201 |
| Toxicity to soil dwelling organisms | : | Method: OECD Test Guideline 217
Remarks: No significant adverse effect on Carbon mineralization.

Method: OECD Test Guideline 216
Remarks: No significant adverse effect on Nitrogen mineralization.

LC50 (Eisenia fetida (earthworms)): > 1.000 mg/kg
Exposure time: 14 d
Method: OECD Test Guideline 207 |
| Toxicity to terrestrial organisms | : | LD50 (Apis mellifera (bees)): 1277,31 µg/bee
Exposure time: 48 h
End point: Acute contact toxicity
Method: OECD Test Guideline 214

LD50 (Apis mellifera (bees)): 728,62 µg/bee
Exposure time: 48 h
End point: Acute oral toxicity
Method: OECD Test Guideline 213

LD50 (Coturnix japonica (Japanese quail)): > 2.000 mg/kg
End point: Acute oral toxicity
Method: US EPA Test Guideline OPPTS 850.2100 |

Components:**chlorothalonil (ISO):**

- | | | |
|------------------|---|---|
| Toxicity to fish | : | LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,059 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 0,039 mg/l
Exposure time: 96 h |
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	Test Type: static test
	LC50 (Pimephales promelas (fathead minnow)): 0,023 mg/l
	Exposure time: 96 h
	Test Type: static test
Toxicity to daphnia and other aquatic invertebrates	: EC50 (Daphnia): 0,07 mg/l
	Exposure time: 48 h
	Test Type: static test
	LC50 (Penaeus duorarum (Pink shrimp)): 165
	Exposure time: 96 h
Toxicity to algae/aquatic plants	: NOEC (Selenastrum capricornutum (green algae)): 0,1 mg/l
	Exposure time: 120 h
	NOEC (Selenastrum capricornutum (green algae)): 0,05 mg/l
	Exposure time: 72 h
	EC50 (Selenastrum capricornutum (green algae)): 0,2 mg/l
	Exposure time: 120 h
	EC50 (Selenastrum capricornutum (green algae)): 0,13 mg/l
	Exposure time: 72 h
	EC50 (Navicula pelliculosa (Freshwater diatom)): 0,0051 mg/l
	Exposure time: 72 h
	EC50 (Lemna gibba (duckweed)): 0,51 mg/l
	Exposure time: 14 d
M-Factor (Acute aquatic toxicity)	: 100
M-Factor (Chronic aquatic toxicity)	: 10
Toxicity to soil dwelling organisms	: LC50 (Eisenia fetida (earthworms)): > 404 mg/kg
	Exposure time: 14 d
Toxicity to terrestrial organisms	: LD50 (Apis mellifera (bees)): > 101 µg/bee
	End point: Acute contact toxicity
	LD50 (Anas platyrhynchos (Mallard duck)): > 4.640 mg/kg
	LD50 (Apis mellifera (bees)): > 63 µg/bee
	End point: Acute oral toxicity
	LC50 (Colinus virginianus (Bobwhite quail)): > 10.000 mg/kg
	Exposure time: 5 d
	LC50 (Anas platyrhynchos (Mallard duck)): > 10.000 mg/kg
	Exposure time: 5 d
	LD50 (Colinus virginianus (Bobwhite quail)): > 2.000 mg/kg

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Fluindapyr:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 0,121 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Oryzias latipes (Japanese medaka)): > 1,8 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Danio rerio (zebra fish)): 0,424 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Cyprinodon variegatus (sheepshead minnow)): 0,43 mg/l
Exposure time: 96 h
Test Type: static test
Method: OPPTS 850.1075
GLP: yes

LC50 (Cyprinus carpio (Carp)): 0,11 mg/l
Exposure time: 96 h
Test Type: Static renewal test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,286 mg/l
Exposure time: 96 h
Test Type: static test
Method: OECD Test Guideline 203
GLP: yes

LC50 (Pimephales promelas (fathead minnow)): 0,19 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0,141 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

LC50 (Americamysis bahia (mysid shrimp)): 0,33 mg/l
Exposure time: 96 h
Test Type: static test
Method: OCSP 850.1035
GLP: yes

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Toxicity to algae/aquatic plants : ErC50 (*Pseudokirchneriella subcapitata* (green algae)): > 4,83 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

NOEC (*Lemna gibba* (duckweed)): 2 mg/l
Exposure time: 7 d
Method: OECD Test Guideline 221
GLP: yes

EC50 (*Skeletonema costatum* (Diatom)): > 2 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

M-Factor (Acute aquatic toxicity) : 1

Toxicity to fish (Chronic toxicity) : NOEC (*Pimephales promelas* (fathead minnow)): 0,031 mg/l
Exposure time: 32 d
Test Type: Early-life Stage
Method: OECD Test Guideline 210
GLP: yes

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC (*Americamysis bahia* (mysid shrimp)): 0,062 mg/l
Exposure time: 28 d
Test Type: flow-through test
Method: OPPTS 850.1350
GLP: yes

NOEC (*Daphnia magna* (Water flea)): 0,12 mg/l
End point: reproduction
Exposure time: 21 d
Method: OECD Test Guideline 211
GLP: yes

NOEC (*Hyalella azteca* (Amphipod)): 68 mg/l
Exposure time: 42 d
Test Type: Static renewal test
GLP: yes

M-Factor (Chronic aquatic toxicity) : 1

Toxicity to soil dwelling organisms : LC50 (*Eisenia fetida* (earthworms)): > 1.000 mg/kg

Method: OECD Test Guideline 216
Remarks: No significant adverse effect on Nitrogen mineralization.

Method: OECD Test Guideline 217
Remarks: No significant adverse effect on Carbon mineraliza-

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tion.

Toxicity to terrestrial organisms : LD50 (Colinus virginianus (Bobwhite quail)): 1.612 mg/kg
Method: OECD Test Guideline 205
GLP: yes

LD50 (Colinus virginianus (Bobwhite quail)): 2.250 mg/kg
Method: OPPTS 850.2100
GLP: yes

LD50 (Apis mellifera (bees)): > 300 µg/bee
Exposure time: 48 h
Method: OECD Test Guideline 214
GLP: yes
Remarks: Contact

LD50 (Apis mellifera (bees)): > 32,8 µg/bee
Exposure time: 48 h
Method: OECD Test Guideline 213
GLP: yes
Remarks: Oral

NOEC (Anas platyrhynchos (Mallard duck)): 679 mg/kg
End point: Reproduction Test
Method: OECD Test Guideline 206
GLP: yes

NOEC (Colinus virginianus (Bobwhite quail)): 174 mg/kg
End point: Reproduction Test
Method: OECD Test Guideline 206
GLP: yes

1,2-benzisothiazol-3(2H)-one:

Toxicity to fish : LC50 (Cyprinodon variegatus (sheepshead minnow)): 16,7 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (Oncorhynchus mykiss (rainbow trout)): 2,15 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 2,9 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 0,070 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,04 mg/l

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Exposure time: 72 h
Method: OECD Test Guideline 201

M-Factor (Acute aquatic toxicity) : 10

Toxicity to microorganisms : EC50 (activated sludge): 24 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

EC50 (activated sludge): 12,8 mg/l
Exposure time: 3 h
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

Persistence and degradability

Components:

chlorothalonil (ISO):

Stability in water : Degradation half life (DT50): 38 d (22 °C) pH: 9

Fluindapyr:

Biodegradability : Result: Not readily biodegradable.

1,2-benzisothiazol-3(2H)-one:

Biodegradability : Result: rapidly biodegradable
Method: OECD Test Guideline 301C

Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

chlorothalonil (ISO):

Bioaccumulation : Remarks: Low potential for bioaccumulation

Partition coefficient: n-octanol/water : log Pow: 2,92

Fluindapyr:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): < 500
Method: OECD Test Guideline 305
GLP: yes
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n- : log Pow: > 3

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octanol/water

1,2-benzisothiazol-3(2H)-one:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Bioconcentration factor (BCF): 6,62
Exposure time: 56 d
Method: OECD Test Guideline 305
Remarks: Substance is not persistent, bioaccumulative, and toxic (PBT).

Partition coefficient: n-octanol/water : log Pow: 0,7 (20 °C)
pH: 7

log Pow: 0,99 (20 °C)
pH: 5

Mobility in soil

Components:

chlorothalonil (ISO):

Distribution among environmental compartments : Koc: 850 - 7000 ml/g
Remarks: Low mobility in soil.

Stability in soil :

Fluindapyr:

Distribution among environmental compartments : Remarks: Low mobility in soil.

1,2-benzisothiazol-3(2H)-one:

Distribution among environmental compartments : Koc: 9,33 ml/g, log Koc: 0,97
Method: OECD Test Guideline 121
Remarks: Highly mobile in soils

Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

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Contaminated packaging : It is prohibited to reuse, bury, burn, or sell containers. Rinsable containers: Triple rinse containers of less than 20 liters and pressure rinse containers of 20 liters or more. Triple rinsing: Add water up to $\frac{1}{4}$ of the container's capacity, close and shake for 30 seconds. Pour the rinse water into the mixing tank, considering this volume of water within the recommended volume for mixing preparation. Perform this procedure three times. Pressure rinsing: Activate the pressure rinsing device for 30 seconds, considering the volume of water used as part of the recommended volume for mixing preparation. In both procedures, punctured the container on its base without damaging the label. In all cases, take the empty containers to collection points indicated by the local empty containers program.

SECTION 14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number	: UN 2810
Proper shipping name	: TOXIC LIQUID, ORGANIC, N.O.S. (chlorothalonil, Fluindapyr)
Class	: 6.1
Packing group	: II
Labels	: 6.1
Environmentally hazardous	: yes

IATA-DGR

UN/ID No.	: UN 2810
Proper shipping name	: Toxic liquid, organic, n.o.s. (chlorothalonil, Fluindapyr)
Class	: 6.1
Packing group	: II
Labels	: Toxic
Packing instruction (cargo aircraft)	: 662
Packing instruction (passenger aircraft)	: 654
Environmentally hazardous	: yes

IMDG-Code

UN number	: UN 2810
Proper shipping name	: TOXIC LIQUID, ORGANIC, N.O.S. (chlorothalonil, Fluindapyr)
Class	: 6.1
Packing group	: II
Labels	: 6.1
EmS Code	: F-A, S-A
Marine pollutant	: yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

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Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

SECTION 15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Argentina. Carcinogenic Substances and Agents Reg- : Not applicable
istry.

Control of precursors and essential chemicals for the : methanol
preparation of drugs.

The ingredients of this product are reported in the following inventories:

TCSI	: Not in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. chlorothalonil (ISO) Monobutyl ether of polymer of 2-methyloxirane / oxirane Fluindapyr Xanthan gum Sulfurous acid, monosodium salt, reaction products with cresol-formaldehyde-nonylphenol polymer Soybean oil sodium hydroxide Smectite-group minerals
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

SECTION 16. OTHER INFORMATION

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Full text of other abbreviations

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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